

**ANNEX III - COURSE SPECIFICATIONS**  
**Bachelor of Science in Computer Engineering**

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## TECHNICAL COURSES

### MATHEMATICS

|                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| <b>Course Name</b>                      | <b>Calculus 1</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Course Description</b>               | <i>An introductory course covering the core concepts of limit, continuity and differentiability of functions involving one or more variables. This also includes the application of differential calculations in solving problems on optimization, rates of change, related rates, tangents and normals, and approximations; partial differentiation and transcendental curve tracing.</i>                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Number of Units for Lecture</b>      | 3 units                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Number of Contact Hours per Week</b> | 3 hours per week                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Prerequisites</b>                    | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Program Outcomes</b>                 | To be identified by the program.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Course Outcomes</b>                  | To be identified by the program.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Course Outline</b>                   | <ol style="list-style-type: none"><li>1. Functions</li><li>2. Continuity and Limits</li><li>3. The Derivative</li><li>4. The Slope</li><li>5. Rate of Change</li><li>6. The Chain Rule and the General Power Rule</li><li>7. Implicit Differentiation</li><li>8. Higher-Order Derivatives</li><li>9. Polynomial Curves</li><li>10. Applications of the Derivative</li><li>11. The Differential</li><li>12. Derivatives of Trigonometric Functions</li><li>13. Derivatives of Inverse Trigonometric Functions</li><li>14. Derivatives of Logarithmic and Exponential Functions</li><li>15. Derivatives of the Hyperbolic Functions</li><li>16. Solutions of Equations</li><li>17. Transcendental Curve Tracing</li><li>18. Parametric Equations</li><li>19. Partial Differentiation</li></ol> |

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| <b>Course Name</b>        | <b>Calculus 2</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Course Description</b> | <i>The course introduces the concept of integration and its application to some physical problems such as evaluation of areas, volumes of revolution, force, and work. The fundamental formulas and various techniques of integration are taken up and applied to both single variable and multi-variable functions. The course also includes tracing of functions of two variables for a better appreciation of the interpretation of the double and triple integral as volume of a three-dimensional region bounded by two or more surfaces.</i> |

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| <b>Number of Units for Lecture</b>      | 3 units                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Number of Contact Hours per Week</b> | 3 hours per week                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Prerequisites</b>                    | Calculus 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Program Outcomes</b>                 | To be identified by the program.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Course Outcomes</b>                  | To be identified by the program.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Course Outline</b>                   | <ol style="list-style-type: none"> <li>1. Integration Concepts/Formulas <ol style="list-style-type: none"> <li>1.1. Anti-Differentiation</li> <li>1.2. Indefinite Integrals</li> <li>1.3. Simple Power Formula</li> <li>1.4. Simple Trigonometric Functions</li> <li>1.5. Logarithmic Function</li> <li>1.6. Exponential Function</li> <li>1.7. Inverse Trigonometric Functions</li> <li>1.8. Hyperbolic Functions (<math>\sinh u</math> &amp; <math>\cosh u</math> only)</li> <li>1.9. General Power formula (include Substitution Rule)</li> <li>1.10. Constant of Integration</li> <li>1.11. Definite Integral (include absolute, odd &amp; even functions)</li> </ol> </li> <li>2. Integration Techniques <ol style="list-style-type: none"> <li>2.1. Integration by Parts</li> <li>2.2. Trigonometric Integrals</li> <li>2.3. Trigonometric Substitution</li> <li>2.4. Rational Functions</li> <li>2.5. Rationalizing Substitution</li> </ol> </li> <li>3. Improper Integrals</li> <li>4. Application of Definite Integral <ol style="list-style-type: none"> <li>4.1. Plane Area</li> <li>4.2. Areas Between Curves</li> </ol> </li> <li>5. Other Applications <ol style="list-style-type: none"> <li>5.1. Volumes</li> <li>5.2. Work</li> <li>5.3. Hydrostatic Pressure</li> </ol> </li> <li>6. Multiple Integrals (Inversion of order/ change of coordinates) <ol style="list-style-type: none"> <li>6.1. Double Integrals</li> <li>6.2. Triple Integrals</li> </ol> </li> <li>7. Surface Tracing <ol style="list-style-type: none"> <li>7.1. Planes</li> <li>7.2. Spheres</li> <li>7.3. Cylinders</li> <li>7.4. Quadric Surfaces</li> <li>7.5. Intersection of Surfaces</li> </ol> </li> <li>8. Multiple Integrals as Volume <ol style="list-style-type: none"> <li>8.1. Double Integrals</li> <li>8.2. Triple Integrals</li> </ol> </li> </ol> |

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| <b>Course Name</b>                      | <b>Engineering Data and Analysis</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Course Description</b>               | <i>This course is designed for undergraduate engineering students with emphasis on problem solving related to societal issues that engineers and scientists are called upon to solve. It introduces different methods of data collection and the suitability of using a particular method for a given situation. The relationship of probability to statistics is also discussed, providing students with the tools they need to understand how "chance" plays a role in statistical analysis. Probability distributions of random variables and their uses are also considered, along with a discussion of linear functions of random variables within the context of their application to data analysis and inference. The course also includes estimation techniques for unknown parameters; and hypothesis testing used in making inferences from sample to population; inference for regression parameters and build models for estimating means and predicting future values of key variables under study. Finally, statistically based experimental design techniques and analysis of outcomes of experiments are discussed with the aid of statistical software.</i>                                                                          |
| <b>Number of Units for Lecture</b>      | 3 units                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Number of Contact Hours per Week</b> | 3 hours per week                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Prerequisites</b>                    | Calculus 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Program Outcomes</b>                 | To be identified by the program.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Course Outcomes</b>                  | To be identified by the program.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Course Outline</b>                   | <ol style="list-style-type: none"> <li>1. Obtaining Data <ol style="list-style-type: none"> <li>1.1. Methods of Data Collection</li> <li>1.2. Planning and Conducting Surveys</li> <li>1.3. Planning and Conducting Experiments: Introduction to Design of Experiments</li> </ol> </li> <li>2. Probability <ol style="list-style-type: none"> <li>2.1. Sample Space and Relationships among Events</li> <li>2.2. Counting Rules Useful in Probability</li> <li>2.3. Rules of Probability</li> </ol> </li> <li>3. Discrete Probability Distributions <ol style="list-style-type: none"> <li>3.1. Random Variables and their Probability Distributions</li> <li>3.2. Cumulative Distribution Functions</li> <li>3.3. Expected Values of Random Variables</li> <li>3.4. The Binomial Distribution</li> <li>3.5. The Poisson Distribution</li> </ol> </li> <li>4. Continuous Probability Distribution <ol style="list-style-type: none"> <li>4.1. Continuous Random Variables and their Probability Distribution</li> <li>4.2. Expected Values of Continuous Random Variables</li> <li>4.3. Normal Distribution</li> <li>4.4. Normal Approximation to the Binomial and Poisson Distribution</li> <li>4.5. Exponential Distribution</li> </ol> </li> </ol> |

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|  | <ul style="list-style-type: none"> <li>5. <i>Joint Probability Distribution</i> <ul style="list-style-type: none"> <li>5.1. <i>Two or Random Variables</i> <ul style="list-style-type: none"> <li>5.1.1. <i>Joint Probability Distributions</i></li> <li>5.1.2. <i>Marginal Probability Distribution</i></li> <li>5.1.3. <i>Conditional Probability Distribution</i></li> <li>5.1.4. <i>More than Two Random Variables</i></li> </ul> </li> <li>5.2. <i>Linear Functions of Random Variables</i></li> <li>5.3. <i>General Functions of Random Variables</i></li> </ul> </li> <li>6. <i>Sampling Distributions and Point Estimation of Parameters</i> <ul style="list-style-type: none"> <li>6.1. <i>Point Estimation</i></li> <li>6.2. <i>Sampling Distribution and the Central Limit Theorem</i></li> <li>6.3. <i>General Concept of Point Estimation</i> <ul style="list-style-type: none"> <li>6.3.1. <i>Unbiased Estimator</i></li> <li>6.3.2. <i>Variance of a Point Estimator</i></li> <li>6.3.3. <i>Standard Error</i></li> <li>6.3.4. <i>Mean Squared Error of an Estimator</i></li> </ul> </li> </ul> </li> <li>7. <i>Statistical Intervals</i> <ul style="list-style-type: none"> <li>7.1. <i>Confidence Intervals: Single Sample</i></li> <li>7.2. <i>Confidence Intervals: Multiple Samples</i></li> <li>7.3. <i>Prediction Intervals</i></li> <li>7.4. <i>Tolerance Intervals</i></li> </ul> </li> <li>8. <i>Test of Hypothesis for a Single Sample</i> <ul style="list-style-type: none"> <li>8.1. <i>Hypothesis Testing</i> <ul style="list-style-type: none"> <li>8.1.1. <i>One-sided and Two-sided Hypothesis</i></li> <li>8.1.2. <i>P-value in Hypothesis Tests</i></li> <li>8.1.3. <i>General Procedure for Test of Hypothesis</i></li> </ul> </li> <li>8.2. <i>Test on the Mean of a Normal Distribution, Variance Known</i></li> <li>8.3. <i>Test on the Mean of a Normal Distribution, Variance Unknown</i></li> <li>8.4. <i>Test on the Variance and Statistical Deviation of a Normal Distribution</i></li> <li>8.5. <i>Test on a Population Proportion</i></li> </ul> </li> <li>9. <i>Statistical Inference of Two Samples</i> <ul style="list-style-type: none"> <li>9.1. <i>Inference on the Difference in Means of Two Normal Distributions, Variances Known</i></li> <li>9.2. <i>Inference on the Difference in Means of Two Normal Distributions, Variances Unknown</i></li> <li>9.3. <i>Inference on the Variance of Two Normal Distributions</i></li> <li>9.4. <i>Inference on Two Population Proportions</i></li> </ul> </li> <li>10. <i>Simple Linear Regression and Correlation</i> <ul style="list-style-type: none"> <li>10.1. <i>Empirical Models</i></li> <li>10.2. <i>Regression: Modelling Linear Relationships – The Least-Squares Approach</i></li> <li>10.3. <i>Correlation: Estimating the Strength of Linear Relation</i></li> <li>10.4. <i>Hypothesis Tests in Simple Linear Regression</i> <ul style="list-style-type: none"> <li>10.4.1. <i>Use of t-tests</i></li> <li>10.4.2. <i>Analysis of Variance Approach to Test Significance of Regression</i></li> </ul> </li> <li>10.5. <i>Prediction of New Observations</i></li> <li>10.6. <i>Adequacy of the Regression Model</i> <ul style="list-style-type: none"> <li>10.6.1. <i>Residual Analysis</i></li> </ul> </li> </ul> </li> </ul> |
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|  | 10.6.2. Coefficient of Determination<br>10.7. Correlation |
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| <b>Course Name</b>                      | <b>Differential Equations</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Course Description</b>               | <i>This course is intended for all engineering students to have a firm foundation on differential equations in preparation for their degree-specific advanced mathematics courses. It covers first order differential equations, nth order linear differential equations and systems of first order linear differential equations. It also introduces the concept of Laplace Transforms in solving differential equations. The students are expected to be able to recognize different kinds of differential equations, determine the existence and uniqueness of solution, select the appropriate methods of solution and interpret the obtained solution. Students are also expected to relate differential equations to various practical engineering and scientific problems as well as employ computer technology in solving and verifying solutions.</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Number of Units for Lecture</b>      | 3 units                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Number of Contact Hours per Week</b> | 3 hours per week                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Prerequisites</b>                    | Calculus 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Program Outcomes</b>                 | To be identified by the program.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Course Outcomes</b>                  | To be identified by the program.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Course Outline</b>                   | <ol style="list-style-type: none"> <li>1. Introduction / Definition <ol style="list-style-type: none"> <li>1.1. Definition and Classifications of Differential Equations (DE)</li> <li>1.2. Solution of a DE</li> </ol> </li> <li>2. Solution of some 1<sup>st</sup> order DE <ol style="list-style-type: none"> <li>2.1. Variable Separable</li> <li>2.2. Exact Equation</li> <li>2.3. Linear Equation</li> <li>2.4. Substitution Methods <ol style="list-style-type: none"> <li>2.4.1. Homogeneous Coefficients</li> <li>2.4.2. Bernoulli's Equation</li> <li>2.4.3. Other Substitution Methods</li> </ol> </li> <li>2.5. Mixed Problems (method not pre-identified)</li> <li>2.6. Introduction to Use of Computer in Solving Differential Equations</li> </ol> </li> <li>3. Application of 1<sup>st</sup> Order Differential Equations <ol style="list-style-type: none"> <li>3.1. Decomposition /Growth</li> <li>3.2. Newton's Law of Cooling</li> <li>3.3. Mixing (non-reacting fluids)</li> <li>3.4. Electric Circuits</li> </ol> </li> <li>4. Linear Differential Equation of Order n <ol style="list-style-type: none"> <li>4.1. Introduction <ol style="list-style-type: none"> <li>4.1.1. Standard form of a n<sup>th</sup> order Linear DE</li> <li>4.1.2. Differential Operators</li> <li>4.1.3. Principle of Superposition</li> </ol> </li> </ol> </li> </ol> |

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|  | <p>4.1.4. Linear Independence of a Set of Functions</p> <p>4.2. Homogeneous Linear Differential Equation with Constant Coefficients</p> <p>4.2.1. Solution of a Homogeneous Linear Ordinary DE</p> <p>4.2.2. Initial and Boundary Value Problems</p> <p>4.3. Non-homogeneous Differential Equation With Constant Coefficients</p> <p>4.3.1. Form of the General Solution</p> <p>4.3.2. Solution by Method of Undetermined Coefficients</p> <p>4.3.3. Solution by Variation of Parameters</p> <p>4.3.4. Mixed Problems</p> <p>4.4. Solution of Higher Order Differential Equations using Computer</p> <p>5. Laplace Transforms of Functions</p> <p>5.1. Definition</p> <p>5.2. Transform of Elementary Functions</p> <p>5.3. Transform of <math>e^{at}f(t)</math> – Theorem</p> <p>5.4. Transform of <math>t^n f(t)</math> – Derivatives of Transforms</p> <p>5.5. Inverse Transforms</p> <p>5.6. Laplace and Inverse Laplace Transforms using a Computer</p> <p>5.7. Transforms of Derivatives</p> <p>5.8. Initial Value Problems</p> <p>6. The Heaviside Unit-Step Function</p> <p>6.1. Definition</p> <p>6.2. Laplace Transforms of Discontinuous Functions and Inverse Transform Leading to Discontinuous Functions</p> <p>6.3. Solution of Initial Value Problems with Discontinuous Functions by Laplace Transform Method</p> <p>7. Application of Laplace Transforms (Problems on Vibration)</p> <p>8. Solution of Systems of Linear Differential Equation with Initial Values/Simultaneous Solution to DE (Laplace Transform Method)</p> |
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#### **NATURAL/PHYSICAL SCIENCES**

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| <b>Course Name</b>                      | <b>Chemistry for Engineers</b>                                                                                                     |
| <b>Course Description</b>               | <i>This course provides students with core concepts of chemistry that are important in the practice of engineering profession.</i> |
| <b>Number of Units for Lecture</b>      | 3 units                                                                                                                            |
| <b>Number of Contact Hours per Week</b> | 3 hours per week                                                                                                                   |
| <b>Prerequisites</b>                    | None                                                                                                                               |
| <b>Co-requisites</b>                    | Chemistry for Engineers Laboratory                                                                                                 |
| <b>Program Outcomes</b>                 | To be identified by the program.                                                                                                   |



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| <b>Course Outcomes</b> | <i>To be identified by the program.</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Course Outline</b>  | <ol style="list-style-type: none"> <li>1. <i>Energy</i> <ol style="list-style-type: none"> <li>1.1 <i>Electrochemical energy</i></li> <li>1.2 <i>Nuclear chemistry and energy</i></li> <li>1.3 <i>Fuels</i></li> </ol> </li> <li>2. <i>The Chemistry of Engineering Materials</i></li> <li>3. <i>Basic Concepts of Crystal Structure</i> <ol style="list-style-type: none"> <li>3.1 <i>Metals</i></li> <li>3.2 <i>Polymers</i></li> <li>3.3 <i>Engineered Nanomaterials</i></li> </ol> </li> <li>4. <i>The Chemistry of the Environment</i></li> <li>5. <i>The Chemistry of the atmosphere</i> <ol style="list-style-type: none"> <li>5.1 <i>The Chemistry of Water</i></li> <li>5.2 <i>Soil chemistry</i></li> </ol> </li> <li>6. <i>Chemical Safety</i></li> <li>7. <i>Special topics specific to field of expertise</i></li> </ol> |

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| <b>Course Name</b>                      | <b>Chemistry for Engineers Laboratory</b>                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Course Description</b>               | <i>A fundamental laboratory course designed to provide opportunity to observe and apply the principles and theories taught in the chemistry for engineers.</i>                                                                                                                                                                                                                                                             |
| <b>Number of Units for Laboratory</b>   | <i>1 unit</i>                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Number of Contact Hours per Week</b> | <i>3 hours per week</i>                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Prerequisites</b>                    | <i>None</i>                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Co-requisites</b>                    | <i>Chemistry for Engineers</i>                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Program Outcomes</b>                 | <i>a-l, b-l, k-l</i>                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Course Outline</b>                   | <ol style="list-style-type: none"> <li>1. <i>Rules and Regulations in the Chemistry Laboratory</i></li> <li>2. <i>Safety Precautions in the Chemistry Laboratory</i></li> <li>3. <i>Rules in Performing an Laboratory Experiment</i></li> <li>4. <i>Making Preliminary and Final Reports</i></li> <li>5. <i>Familiarization of Equipment/Apparatus</i></li> <li>6. <i>Performance of Laboratory Experiments</i></li> </ol> |
| <b>Laboratory Experiments</b>           | <i>Laboratory exercises to be identified by the program.</i><br><i>Each major topic should have a corresponding laboratory exercise.</i><br><i>For semestral program, 15 exercises per semester.</i><br><i>For trimestral program, 12 exercises per trimester.</i><br><i>For quarterm program, 9 exercises per quarter.</i>                                                                                                |

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| <b>Laboratory Equipment</b> | See Annex of Lab Requirements |
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| <b>Course Name</b>                      | <b>Physics for Engineers</b>                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Course Description</b>               | <i>This course covers vectors; kinematics; dynamics; work, energy, and power; impulse and momentum; rotation; dynamics of rotation; elasticity; and oscillation. Fluids; thermal expansion, thermal stress; heat transfer; calorimetry; waves; electrostatics; electricity; magnetism; optics; image formation by plane and curved mirrors; and image formation by thin lenses.</i>                                             |
| <b>Number of Units for Lecture</b>      | 3 units                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Number of Contact Hours per Week</b> | 3 hours per week                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Prerequisites</b>                    | Calculus 1                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Co-requisites</b>                    | Physics for Engineers Laboratory                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Program Outcomes</b>                 | To be identified by the program.                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Course Outcomes</b>                  | To be identified by the program.                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Course Outline</b>                   | <ol style="list-style-type: none"> <li>1. Work, Energy and Power</li> <li>2. Impulse and Momentum</li> <li>3. Kinematics</li> <li>4. Dynamics</li> <li>5. Rotation</li> <li>6. Dynamics of Rotation</li> <li>7. Elasticity</li> <li>8. Oscillations</li> <li>9. Fluids</li> <li>10. Heat Transfer</li> <li>11. Waves</li> <li>12. Electrostatics</li> <li>13. Electricity</li> <li>14. Magnetism</li> <li>15. Optics</li> </ol> |

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| <b>Course Name</b>        | <b>Physics for Engineers Laboratory</b>                                                                                                                      |
| <b>Course Description</b> | <i>A fundamental laboratory course designed to provide opportunity to observe and apply the principles and theories taught in the physics for engineers.</i> |

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| <b>Number of Units for Laboratory</b>   | 1 unit                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Number of Contact Hours per Week</b> | 3 hours per week                                                                                                                                                                                                                                                                                                                                                             |
| <b>Prerequisites</b>                    | None                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Co-requisites</b>                    | Physics for Engineers                                                                                                                                                                                                                                                                                                                                                        |
| <b>Program Outcomes</b>                 | To be identified by the program.                                                                                                                                                                                                                                                                                                                                             |
| <b>Course Outcomes</b>                  | To be identified by the program.                                                                                                                                                                                                                                                                                                                                             |
| <b>Course Outline</b>                   | <ol style="list-style-type: none"> <li>1. Rules and Regulations in the Physics Laboratory</li> <li>2. Safety Precautions in the Physics Laboratory</li> <li>3. Rules in Performing an Laboratory Experiment</li> <li>4. Making Preliminary and Final Reports</li> <li>5. Familiarization of Equipment/Apparatus</li> <li>6. Performance of Laboratory Experiments</li> </ol> |
| <b>Laboratory Experiments</b>           | <p>Laboratory exercises to be identified by the program.</p> <p>Each major topic should have a corresponding laboratory exercise.</p> <p>For semestral program, 15 exercises per semester.</p> <p>For trimestral program, 12 exercises per trimester.</p> <p>For quarterterm program, 9 exercises per quarter.</p>                                                           |
| <b>Laboratory Equipment</b>             | See Annex of Lab Requirements                                                                                                                                                                                                                                                                                                                                                |

### **BASIC ENGINEERING SCIENCE**

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| <b>Course Name</b>                      | <b>Computer Aided Drafting</b>                                                                                                                                                                   |
| <b>Course Description</b>               | Concepts of computer-aided drafting (CAD); introduction to the CAD environment; terminologies; and the general operating procedures and techniques in entering and executing basic CAD commands. |
| <b>Number of Units Laboratory</b>       | 1 unit                                                                                                                                                                                           |
| <b>Number of Contact Hours per Week</b> | 3 hours per week                                                                                                                                                                                 |
| <b>Prerequisites</b>                    | 2 <sup>nd</sup> Year Standing                                                                                                                                                                    |
| <b>Program Outcomes</b>                 | To be identified by the program.                                                                                                                                                                 |
| <b>Course Outcomes</b>                  | To be identified by the program.                                                                                                                                                                 |
| <b>Course Outline</b>                   | <ol style="list-style-type: none"> <li>1. Introduction to CAD Software</li> </ol>                                                                                                                |

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|                             | 2. CAD Drawing<br>3. Snapping, Construction Elements<br>4. Dimensioning<br>5. Plotting, Inputting Images<br>6. 3D and Navigating in 3D<br>7. Rendering |
| <b>Laboratory Equipment</b> | See Annex of Lab Requirements                                                                                                                          |

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| <b>Course Name</b>                      | <b>Engineering Economics</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Course Description</b>               | <i>This course deals with the study of concepts of the time value of money and equivalence; basic economic study methods; decisions under certainty; decisions recognizing risk; and decisions admitting uncertainty.</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Number of Units Lecture</b>          | 3 units                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Number of Contact Hours per Week</b> | 3 hours per week                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Prerequisites</b>                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Program Outcomes</b>                 | e-E, k-E                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Course Outcomes</b>                  | After completing this course, the student must be able to: <ol style="list-style-type: none"> <li>1. Solve problems involving interest and the time value of money;</li> <li>2. Evaluate project alternatives by applying engineering economic principles and methods and select the most economically efficient one; and</li> <li>3. Deal with risk and uncertainty in project outcomes by applying the basic economic decision making concepts.</li> </ol>                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Course Outline</b>                   | <ol style="list-style-type: none"> <li>1. Introduction <ol style="list-style-type: none"> <li>1.1. Definitions</li> <li>1.2. Principles of Engineering Economics</li> <li>1.3. Engineering Economics and the Design Process</li> <li>1.4. Cost Concepts for Decision Making</li> <li>1.5. Present Economic Studies</li> </ol> </li> <li>2. Money-Time Relationships and Equivalence <ol style="list-style-type: none"> <li>2.1. Interest and the Time Value of Money</li> <li>2.2. The Concept of Equivalence</li> <li>2.3. Cash Flows</li> </ol> </li> <li>3. Economic Study Methods <ol style="list-style-type: none"> <li>3.1. The Minimum Attractive Rate of Return</li> <li>3.2. Basic Economic Study Methods: Present Worth, Future Worth, Annual Worth, Internal Rate of Return, External Rate of Return</li> </ol> </li> </ol> |

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|  | <ul style="list-style-type: none"> <li>3.3. <i>Other Methods: Discounted Payback Period, Benefit/Cost Ratio</i></li> <li>4. <i>Decisions Under Certainty</i> <ul style="list-style-type: none"> <li>4.1. <i>Evaluation of Mutually Exclusive Alternatives</i></li> <li>4.2. <i>Evaluation of Independent Projects</i></li> <li>4.3. <i>Effects of Inflation</i></li> <li>4.4. <i>Depreciation and After-Tax Economic Analysis</i></li> <li>4.5. <i>Replacement Studies</i></li> </ul> </li> <li>5. <i>Decisions Recognizing Risk</i> <ul style="list-style-type: none"> <li>5.1. <i>Expected Monetary Value of Alternatives</i></li> <li>5.2. <i>Discounted Decision Tree Analysis</i></li> </ul> </li> <li>6. <i>Decisions Admitting Uncertainty</i> <ul style="list-style-type: none"> <li>6.1. <i>Sensitivity Analysis</i></li> <li>6.2. <i>Decision Analysis Models</i></li> </ul> </li> </ul> |
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| <b>Course Name</b>                      | <b>Engineering Management</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Course Description</b>               | <i>Decision-making; the functions of management; managing production and service operations; managing the marketing function; and managing the finance function.</i>                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Number of Units Lecture</b>          | <i>2 units</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Number of Contact Hours per Week</b> | <i>2 hours per week</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Prerequisites</b>                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Program Outcomes</b>                 | <i>i – I, I - I</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Course Outcomes</b>                  | <i>After completing this course, the student must be able to:</i> <ul style="list-style-type: none"> <li>1. <i>Explain the basic concepts of engineering management;</i></li> <li>2. <i>Apply the principles of engineering management to analyze case studies.</i></li> </ul>                                                                                                                                                                                                                                                                                                       |
| <b>Course Outline</b>                   | <ul style="list-style-type: none"> <li>1. <i>Introduction to Engineering Management</i></li> <li>2. <i>Decision Making</i></li> <li>3. <i>Functions of Management</i> <ul style="list-style-type: none"> <li>3.1. <i>Planning / Coordinating</i></li> <li>3.2. <i>Organizing</i></li> <li>3.3. <i>Staffing</i></li> <li>3.4. <i>Communicating</i></li> <li>3.5. <i>Motivating</i></li> <li>3.6. <i>Leading</i></li> <li>3.7. <i>Controlling</i></li> </ul> </li> <li>4. <i>Managing Product and Service Operations</i></li> <li>5. <i>Managing the Marketing Function</i></li> </ul> |

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|  | 6. <i>Managing the Finance Function</i> |
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### **ALLIED COURSES**

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| <b>Course Name</b>                      | <b>Fundamentals of Electrical Circuits</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Course Description</b>               | <i>This course introduces the fundamental concepts, circuit laws, theorems and techniques used in electrical circuit analysis and transient analysis, as well as its application. The course covers circuit topologies and DC excitations, transient response, AC response, and polyphase circuits. The use of computer software for circuit simulation and design are emphasized to expose students to computer-based tools.</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Number of Units for Lecture</b>      | 3 units                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Number of Contact Hours per Week</b> | 3 hours per week                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Prerequisites</b>                    | Physics for Engineers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Co-requisites</b>                    | Fundamentals of Electrical Circuits Laboratory                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Program Outcomes</b>                 | To be identified by the program.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Course Outcomes</b>                  | To be identified by the program.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Course Outline</b>                   | <ol style="list-style-type: none"> <li>1. <i>Circuit Topologies and DC Excitations</i> <ol style="list-style-type: none"> <li>1.1 <i>Introductory Concepts</i></li> <li>1.2 <i>Electrical Properties of Materials</i></li> <li>1.3 <i>Passive Elements</i></li> <li>1.4 <i>Network Laws and Theorems</i></li> <li>1.5 <i>Electric Circuit Theorems</i></li> </ol> </li> <li>2. <i>Transient Response</i> <ol style="list-style-type: none"> <li>2.1 <i>RC Circuits</i></li> <li>2.2 <i>RL Circuits</i></li> <li>2.3 <i>RLC Circuits</i></li> </ol> </li> <li>3. <i>AC Response and Polyphase Circuits</i> <ol style="list-style-type: none"> <li>3.1 <i>Reactance and Impedance</i></li> <li>3.2 <i>Introduction to Single-Phase AC</i></li> <li>3.3 <i>AC Power Analysis</i></li> <li>3.4 <i>Sinusoidal Steady-State Analysis</i></li> <li>3.5 <i>Three-Phase Circuits</i></li> <li>3.6 <i>Transformer</i></li> </ol> </li> </ol> |

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| <b>Course Name</b>        | <b>Fundamentals of Electrical Circuits Laboratory</b>                                                                                     |
| <b>Course Description</b> | <i>This course allows the students to verify the laws and theorems discussed in Fundamentals of Electrical Circuits (lecture) through</i> |

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|                                         | <i>simulation, experimentation and project construction. The course topics include experimental determination of the characteristics of the different circuit configurations (series, parallel, series/parallel, delta, and wye), electrical power, Ohm's Law, Kirchhoff's Voltage and Current Laws, Superposition Theorem, Thevenin's equivalent circuit, and maximum power transfer. The use of computer software for circuit simulation and design are used as basis in verifying experimental results and to expose students to computer-based tools.</i> |
| <b>Number of Units for Laboratory</b>   | <i>1 unit</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Number of Contact Hours per Week</b> | <i>3 hours per week</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Prerequisites</b>                    | <i>Physics for Engineers</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Co-requisites</b>                    | <i>Fundamentals of Electrical Circuits</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Program Outcomes</b>                 | <i>To be identified by the program.</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Course Outcomes</b>                  | <i>To be identified by the program.</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Course Outline</b>                   | <ol style="list-style-type: none"> <li><i>1. Rules and Regulations in the Electrical Engineering Laboratory</i></li> <li><i>2. Safety Precautions in the Electrical Engineering Laboratory</i></li> <li><i>3. Rules in Performing an Laboratory Experiment</i></li> <li><i>4. Making Preliminary and Final Reports</i></li> <li><i>5. Familiarization of Equipment/Apparatus</i></li> <li><i>6. Performance of Laboratory Experiments</i></li> </ol>                                                                                                          |
| <b>Laboratory Experiments</b>           | <i>Laboratory exercises to be identified by the program.</i><br><i>Each major topic should have a corresponding laboratory exercise.</i><br><i>For semestral program, 15 exercises per semester.</i><br><i>For trimestral program, 12 exercises per trimester.</i><br><i>For quarterterm program, 9 exercises per quarter.</i>                                                                                                                                                                                                                                |
| <b>Laboratory Equipment</b>             | <i>Program shall provide complete tools and equipment necessary to perform the identified laboratory exercise.</i><br><i>1 set of tools and equipment per maximum of 5 students per group.</i>                                                                                                                                                                                                                                                                                                                                                                |

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| <b>Course Name</b>                 | <b>Fundamentals of Electronic Circuits</b>                                                                                                                                                                                                 |
| <b>Course Description</b>          | <i>This course discusses the construction, operation and characteristics of basic electronic devices such as junction diodes, bipolar junction transistors, Field Effect Transistors and MOS Field Effect Transistors and oscillators.</i> |
| <b>Number of Units for Lecture</b> | <i>3 units</i>                                                                                                                                                                                                                             |

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| <b>Number of Contact Hours per Week</b> | 3 hours per week                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Prerequisites</b>                    | Fundamentals of Electrical Circuits                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Co-requisites</b>                    | Fundamentals of Electronic Circuits Laboratory                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Program Outcomes</b>                 | To be identified by the program.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Course Outcomes</b>                  | To be identified by the program.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Course Outline</b>                   | <ol style="list-style-type: none"> <li>1. Introduction to Electronics</li> <li>2. Solid State Fundamentals</li> <li>3. Semiconductor PN Junction Diode</li> <li>4. Diode Circuit Analysis and Applications</li> <li>5. DC Regulated Power Supply</li> <li>6. IC Regulator</li> <li>7. Light Emitting Diode (LED)</li> <li>8. Bipolar Junction Transistor (BJT)</li> <li>9. Field Effect Transistor (FET)</li> <li>10. Switching Transistor Circuits</li> <li>11. Resistor Transistor Logic (RTL)</li> <li>12. Direct-Coupled Transistor Logic (DCTL)</li> <li>13. Integrated-Injection Logic (I<sup>2</sup>L)</li> <li>14. Schottky-Diode Non-Saturating Logic (Schottky-Clamped Logic for TTL/I<sup>2</sup>L)</li> <li>15. Emitter-Coupled Logic (ECL)</li> <li>16. MOSFET Logic</li> <li>17. Comparison of Different Logic Families (Summary)</li> <li>18. Oscillator Circuits</li> </ol> |

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| <b>Course Name</b>                      | <b>Fundamentals of Electronic Circuits Laboratory</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Course Description</b>               | This course is the laboratory component of the course Fundamentals of Electronic Circuits (Lecture) that allows students to verify theoretical concepts pertaining to the operation of electronic devices such as the PN junction diodes, BJT and FET and their subsequent applications to electronics circuits involving rectification, amplification and switching applications. The use of laboratory equipment and apparatus to verify the characteristics of diodes and transistor devices, and their operations in circuits such as rectifiers, voltage regulators, amplifiers, oscillators and switches are emphasized. Such equipment includes but not limited to the curve tracer, the oscilloscope, signal generator and multi-meters. |
| <b>Number of Units for Laboratory</b>   | 1 unit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Number of Contact Hours per Week</b> | 3 hours per week                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |



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| <b>Prerequisites</b>          | <i>Fundamentals of Electrical Circuits</i>                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Co-requisites</b>          | <i>Fundamentals of Electronic Circuits</i>                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Program Outcomes</b>       | <i>To be identified by the program.</i>                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Course Outcomes</b>        | <i>To be identified by the program.</i>                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Course Outline</b>         | <ol style="list-style-type: none"> <li>1. <i>Rules and Regulations in the Electronics Engineering Laboratory</i></li> <li>2. <i>Safety Precautions in the Electronics Engineering Laboratory</i></li> <li>3. <i>Rules in Performing an Laboratory Experiment</i></li> <li>4. <i>Making Preliminary and Final Reports</i></li> <li>5. <i>Familiarization of Equipment/Apparatus</i></li> <li>6. <i>Performance of Laboratory Experiments</i></li> </ol> |
| <b>Laboratory Experiments</b> | <p><i>Laboratory exercises to be identified by the program.</i></p> <p><i>Each major topic should have a corresponding laboratory exercise.</i></p> <p><i>For semestral program, 15 exercises per semester.</i></p> <p><i>For trimestral program, 12 exercises per trimester.</i></p> <p><i>For quarterterm program, 9 exercises per quarter.</i></p>                                                                                                  |
| <b>Laboratory Equipment</b>   | <p><i>Program shall provide complete tools and equipment necessary to perform the identified laboratory exercise.</i></p> <p><i>1 set of tools and equipment per maximum of 5 students per group.</i></p>                                                                                                                                                                                                                                              |

### **PROFESSIONAL COURSES**

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| <b>Course Name</b>                      | <b>Computer Engineering as a Discipline</b>                                                                                                                                                                                                                                                        |
| <b>Course Description</b>               | <i>This course discusses the curriculum for Computer Engineering as well as how to prepare students for success through engineering design process, ethical decision-making, teamwork, and communicating to diverse audiences.</i>                                                                 |
| <b>Number of Units for Lecture</b>      | <i>1 unit</i>                                                                                                                                                                                                                                                                                      |
| <b>Number of Contact Hours per Week</b> | <i>1 hour per week</i>                                                                                                                                                                                                                                                                             |
| <b>Prerequisites</b>                    | <i>None</i>                                                                                                                                                                                                                                                                                        |
| <b>Program Outcomes</b>                 | <i>To be identified by the program.</i>                                                                                                                                                                                                                                                            |
| <b>Course Outcomes</b>                  | <i>To be identified by the program.</i>                                                                                                                                                                                                                                                            |
| <b>Course Outline</b>                   | <ol style="list-style-type: none"> <li>1. <i>CpE Curriculum</i></li> <li>2. <i>Introduction to the Engineering Profession</i></li> <li>3. <i>Preparing for an Engineering Career</i></li> <li>4. <i>Introduction to Engineering Design</i></li> <li>5. <i>Engineering Communication</i></li> </ol> |

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|  | 6. Engineering Ethics |
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| <b>Course Name</b>                      | <b>Discrete Mathematics</b>                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Course Description</b>               | <i>This course deals with logic, sets, proofs, growth of functions, theory of numbers, counting techniques, trees and graph theory.</i>                                                                                                                                                                                                                                                                                                                   |
| <b>Number of Units for Lecture</b>      | 3 units                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Number of Contact Hours per Week</b> | 3 hours per week                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Prerequisites</b>                    | Calculus 1                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Program Outcomes</b>                 | To be Identified by the program.                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Course Outcomes</b>                  | To be Identified by the program.                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Course Outline</b>                   | <ol style="list-style-type: none"> <li>1. Logic, Sets, Proofs, and Functions</li> <li>2. Algorithms, Integers and Matrices <ol style="list-style-type: none"> <li>2.1 Growth of Functions</li> <li>2.2 Complexity of Algorithms</li> <li>2.3 Number Theory</li> <li>2.4 Matrices</li> </ol> </li> <li>3. Counting Techniques</li> <li>4. Relations</li> <li>5. Graph Theory</li> <li>6. Trees</li> <li>7. Introduction to Modeling Computation</li> </ol> |

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| <b>Course Name</b>                      | <b>Numerical Methods</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Course Description</b>               | <i>This course covers the concepts of numerical analysis and computer software tools in dealing with engineering problems. It includes techniques in finding the roots of an equation, solving systems of linear and non-linear equations, eigenvalue problems, polynomial approximation and interpolation, ordinary and partial differential equations. The Monte-Carlo method, simulation, error propagation and analysis, the methods of least squares and goodness-of-fit tests are also discussed</i> |
| <b>Number of Units for Lecture</b>      | 3 units                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Number of Contact Hours per Week</b> | 3 hours per week                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Prerequisites</b>                    | Differential Equations                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Program Outcomes</b>                 | To be identified by the program.                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Course Outcomes</b>                  | To be identified by the program.                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Course Outline</b>                   | <ol style="list-style-type: none"> <li>1. Introduction</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                          |

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|  | <ol style="list-style-type: none"> <li>2. Non Linear Transcendental and Polynomial Function Techniques</li> <li>3. Iterative Bracketing Method</li> <li>4. Iterative Non-Bracketing/Open Method</li> <li>5. Iterative Polynomial Function Techniques</li> <li>6. System of Linear Equations</li> <li>7. Direct Methods</li> <li>8. Iterative Methods</li> <li>9. Curve Fitting Techniques</li> <li>10. Least Square Regression</li> <li>11. Interpolation Techniques</li> <li>12. Numerical Integration Techniques</li> <li>13. Numerical Differentiation</li> <li>14. Ordinary Differential Equations</li> </ol> |
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| <b>Course Name</b>                      | <b>Fundamentals of Mixed Signals and Sensors</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Course Description</b>               | <i>This course covers operational amplifiers, signal converters, power switching devices and the construction and operation of sensors and transducers for converting physical parameters into electrical signals and vice-versa. The course focuses on the application of these devices in developing signal conversion circuits that allows measurement, processing and control of physical parameters by digital processing systems such as a finite state machine or a digital computer. Topics on actuators are also included.</i> |
| <b>Number of Units for Lecture</b>      | 3 units                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Number of Contact Hours per Week</b> | 3 hours per week                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Prerequisites</b>                    | Fundamentals of Electronic Circuits                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Program Outcomes</b>                 | To be identified by the program.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Course Outcomes</b>                  | To be identified by the program.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Course Outline</b>                   | <ol style="list-style-type: none"> <li>1. Differential Amplifiers</li> <li>2. Operational Amplifiers</li> <li>3. Linear and Non-Linear Applications of Op-Amp</li> <li>4. Analog to Digital Conversion</li> <li>5. Digital to Analog Conversion</li> <li>6. Other Mixed Signals</li> <li>7. Basic Measurement Theory</li> <li>8. Sensors and Transducers</li> <li>9. Other Types of Sensors and Transducers</li> <li>10. Basic Control Devices</li> </ol>                                                                               |

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| <b>Course Name</b>                      | <b>Programming Logic and Design</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Course Description</b>               | <i>This is an introductory course in computer programming logic. The student will learn algorithms applicable to all programming languages, including: identifiers, data types, arrays, control structures, modular programming, generating reports, and computer memory concepts. The student will learn to use charts commonly used in business and information processing. Program logic will be developed using flowcharts and pseudo code. Programs will be written using any programming language.</i>                                                                                                                              |
| <b>Number of Units for Laboratory</b>   | 2 units                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Number of Contact Hours per Week</b> | 6 hours per week                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Prerequisites</b>                    | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Program Outcomes</b>                 | c – I                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Course Outcomes</b>                  | <p>After completing this course, the student must be able to:</p> <ol style="list-style-type: none"> <li>1. Identify important steps in program development cycle</li> <li>2. Draw a flowchart to represent the program's logic</li> <li>3. Break down programming problems into modules</li> </ol>                                                                                                                                                                                                                                                                                                                                       |
| <b>Course Outline</b>                   | <ol style="list-style-type: none"> <li>1. Introduction to Computers and Logic</li> <li>2. Tools for Developing Program Logic: Flowchart and Pseudocode Instructions</li> <li>3. Logical Control Structures: Sequence, Selection/Decision, Iteration/Loop, Case</li> <li>4. Data and Data Types, Constants and Variables, Operators And Its Hierarchy</li> <li>5. Looping: While Loop, Do...While Loop, For Loop</li> <li>6. Lists and Arrays: Representation, Arrays Interpolation, Add and Delete, Operators and Functions, Slicing</li> <li>7. Debugging Techniques: Steps/Process in Debugging, Approaches, Debugging Tools</li> </ol> |
| <b>Laboratory Experiments</b>           | <p>Laboratory exercises to be identified by the program.<br/> Each major topic should have a corresponding laboratory exercise.<br/> For semestral program, 15 exercises per semester.<br/> For trimestral program, 12 exercises per trimester.<br/> For quarterterm program, 9 exercises per quarter.</p>                                                                                                                                                                                                                                                                                                                                |
| <b>Laboratory Equipment</b>             | <p>Computer and object-oriented programming software tool<br/> Depending on the class size<br/> 1 computer per student</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

Sample Model Course Specification

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| <b>Course Name</b>                      | <b>Object Oriented Programming</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Course Description</b>               | <i>Introduces the fundamental concepts of programming from an object oriented perspective. Topics are drawn from classes and objects, abstraction, encapsulation, data types, calling methods and passing parameters, decisions, loops, arrays and collections, documentation, testing and debugging, exceptions, design issues, inheritance, and polymorphic variables and methods. The course emphasizes modern software engineering and design principles.</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Number of Units for Laboratory</b>   | <i>2 units</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Number of Contact Hours per Week</b> | <i>6 hours per week</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Prerequisites</b>                    | <i>Programming Logic and Design</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Program Outcomes</b>                 | <i>To be identified by the program.</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Course Outcomes</b>                  | <i>To be identified by the program.</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Course Outline</b>                   | <ol style="list-style-type: none"> <li>1. <i>Introduction to Object Oriented Programming and UML</i> <ol style="list-style-type: none"> <li>1.1. <i>Fundamental Concepts: Classes, Objects, and Methods, Inheritance, Encapsulation and Abstraction, Polymorphism</i></li> <li>1.2. <i>Unified Modeling Language (UML): Basic Concepts, Association, Aggregation, Composition, and Multiplicity, UML Diagrams</i></li> </ol> </li> <li>2. <i>Object Oriented Analysis and Design</i> <ol style="list-style-type: none"> <li>2.1. <i>Cohesion and Coupling Concepts</i></li> <li>2.2. <i>Data-Driven Design</i></li> <li>2.3. <i>Responsibility-Driven Design</i></li> <li>2.4. <i>Object-Oriented Design using UML</i></li> </ol> </li> <li>3. <i>Programming Language Fundamentals</i> <ol style="list-style-type: none"> <li>3.1. <i>Coding Conventions and Data Types</i></li> <li>3.2. <i>Constants and Variables</i></li> <li>3.3. <i>Attributes, Methods, and Constructors</i></li> <li>3.4. <i>Control and Iterative Statements</i></li> <li>3.5. <i>Characters and Strings</i></li> <li>3.6. <i>Arrays</i></li> </ol> </li> <li>4. <i>Advanced Programming Language Fundamentals</i> <ol style="list-style-type: none"> <li>4.1. <i>Inheritance</i></li> <li>4.2. <i>Abstract Classes</i></li> </ol> </li> <li>5. <i>Exception Handling</i> <ol style="list-style-type: none"> <li>5.1. <i>Understanding Errors and Exceptions</i></li> <li>5.2. <i>Try, Catch, and Finally</i></li> </ol> </li> <li>6. <i>Graphical User Interface Programming</i> <ol style="list-style-type: none"> <li>6.1. <i>Forms and Widgets</i></li> <li>6.2. <i>Graphics, Images, and Sound</i></li> <li>6.3. <i>Layout Managers</i></li> <li>6.4. <i>Event Handling</i></li> </ol> </li> </ol> |

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| <b>Laboratory Experiments</b> | Laboratory exercises to be identified by the program.<br>Each major topic should have a corresponding laboratory exercise.<br>For semestral program, 15 exercises per semester.<br>For trimestral program, 12 exercises per trimester.<br>For quarterterm program, 9 exercises per quarter. |
| <b>Laboratory Equipment</b>   | Computer and object-oriented programming software tool<br>Depending on the class size<br>1 computer per student                                                                                                                                                                             |

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| <b>Course Name</b>                      | <b>Data Structures and Algorithms</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Course Description</b>               | Solving computational problems that involve manipulating collections of data, study a core set of data abstractions, data structures, and algorithms that provide a foundation for writing efficient programs.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Number of Units for Laboratory</b>   | 2 units                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Number of Contact Hours per Week</b> | 6 hours per week                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Prerequisites</b>                    | Object Oriented Programming                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Program Outcomes</b>                 | To be identified by the program.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Course Outcomes</b>                  | To be identified by the program.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Course Outline</b>                   | <ol style="list-style-type: none"> <li>1. Pointers, Dynamic Memory Allocation, Pointers, Arrays, Structures</li> <li>2. Abstract Data Types (ADT) and Fundamentals of Linked Lists</li> <li>3. Linked Lists Operations</li> <li>4. Stack Abstract Data Type and Its Linked Lists Operations</li> <li>5. Queue Abstract Data Type and Its Linked Lists Operations</li> <li>6. Algorithm Analysis and Linked List Types: Doubly Linked Lists</li> <li>7. Tree ADT and Binary Search Tree</li> <li>8. AVL Tree</li> <li>9. Heaps</li> <li>10. Basic Algorithmic Analysis</li> <li>11. Algorithmic Strategies</li> <li>12. Classic Algorithms For Common Tasks</li> <li>13. Analysis and Design of Application-Specific Algorithms</li> <li>14. Parallel Algorithms and Multithreading</li> <li>15. Algorithmic Complexity</li> <li>16. Scheduling Algorithms</li> <li>17. Basic Computability Theory</li> </ol> |
| <b>Laboratory Experiments</b>           | Laboratory exercises to be identified by the program.<br>Each major topic should have a corresponding laboratory exercise.<br>For semestral program, 15 exercises per semester.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

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|                             | <i>For trimestral program, 12 exercises per trimester.<br/>For quarterterm program, 9 exercises per quarter.</i> |
| <b>Laboratory Equipment</b> | <i>Computer and any programming software tool<br/>Depending on the class size<br/>1 computer per student</i>     |

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| <b>Course Name</b>                      | <b>Software Design</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Course Description</b>               | <i>This course focuses on programming paradigms and constructs, data structures and use of standard library functions for manipulating them, object-oriented design and the use of modeling languages, testing and software quality concepts, and tradeoffs among different software design methods.</i>                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Number of Units for Lecture</b>      | <i>3 units</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Number of Contact Hours per Week</b> | <i>3 hours per week</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Prerequisites</b>                    | <i>Data Structures and Algorithms</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Co-requisites</b>                    | <i>Software Design Laboratory</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Program Outcomes</b>                 | <i>To be identified by the program.</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Course Outcomes</b>                  | <i>To be identified by the program.</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Course Outline</b>                   | <ol style="list-style-type: none"> <li>1. <i>History and Overview</i></li> <li>2. <i>Relevant Tools, Standards, and/or Engineering Constraints</i></li> <li>3. <i>Programming Constructs and Paradigms</i></li> <li>4. <i>Problem-Solving Strategies</i></li> <li>5. <i>Data Structures</i></li> <li>6. <i>Recursion</i></li> <li>7. <i>Object-Oriented Design</i></li> <li>8. <i>Software Testing and Quality</i></li> <li>9. <i>Data Modeling</i></li> <li>10. <i>Database Systems</i></li> <li>11. <i>Event-Driven and Concurrent Programming</i></li> <li>12. <i>Using Application Programming Interfaces</i></li> <li>13. <i>Data Mining</i></li> <li>14. <i>Data Visualization</i></li> </ol> |

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| <b>Course Name</b>        | <b>Software Design Laboratory</b>                                               |
| <b>Course Description</b> | <i>This course focuses on providing hands-on experience in software design.</i> |

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| <b>Number of Units for Lecture</b>      | 1 unit                                                                                                                                                                                                                                                                                      |
| <b>Number of Contact Hours per Week</b> | 3 hours per week                                                                                                                                                                                                                                                                            |
| <b>Prerequisites</b>                    | Data Structures and Algorithms                                                                                                                                                                                                                                                              |
| <b>Co-requisites</b>                    | Software Design                                                                                                                                                                                                                                                                             |
| <b>Program Outcomes</b>                 | To be identified by the program.                                                                                                                                                                                                                                                            |
| <b>Course Outcomes</b>                  | To be identified by the program.                                                                                                                                                                                                                                                            |
| <b>Laboratory Experiments</b>           | Laboratory exercises to be identified by the program.<br>Each major topic should have a corresponding laboratory exercise.<br>For semestral program, 15 exercises per semester.<br>For trimestral program, 12 exercises per trimester.<br>For quarterterm program, 9 exercises per quarter. |
| <b>Laboratory Equipment</b>             | Computer and object-oriented programming software tool<br>Depending on the class size<br>1 computer per student                                                                                                                                                                             |

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| <b>Course Name</b>                      | <b>Logic Circuits and Design</b>                                                                                                                                                                                                                                                                                                   |
| <b>Course Description</b>               | The course includes design and analysis of digital circuits. This course covers both combinational (synchronous and asynchronous) logic circuits with emphasis on solving digital problems using hardwired structures of the complexity of medium and large-scale integration.                                                     |
| <b>Number of Units for Lecture</b>      | 3 units                                                                                                                                                                                                                                                                                                                            |
| <b>Number of Contact Hours per Week</b> | 3 hours per week                                                                                                                                                                                                                                                                                                                   |
| <b>Prerequisites</b>                    | Fundamentals of Electronic Circuits                                                                                                                                                                                                                                                                                                |
| <b>Co-requisites</b>                    | Logic Circuits and Design Laboratory                                                                                                                                                                                                                                                                                               |
| <b>Program Outcomes</b>                 | To be identified by the program.                                                                                                                                                                                                                                                                                                   |
| <b>Course Outcomes</b>                  | To be identified by the program.                                                                                                                                                                                                                                                                                                   |
| <b>Course Outline</b>                   | <ol style="list-style-type: none"> <li>1. Gates, Truth Tables, Boolean Algebra, Function Simplification</li> <li>2. K-Maps, Circuit Implementation Using K-Maps, SOP and POS Representation, NAND/NOR Implementations.</li> <li>3. Mux, Demux, Decoders, Code Conversion (BCD to Binary, Excess-3 to Binary, Gray Code)</li> </ol> |



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|  | 4. Latches and Flip-Flops: SR, D, JK, T<br>5. Counter Design, Register Design, ALU Function<br>6. Sequential Circuits, Excitation Function, State Table, State Diagram.<br>7. Sequential Circuit Design with Different Flip-Flops.<br>8. Synchronous and Asynchronous Circuits Analysis and Design, Excitation Function, Flow Table<br>9. Algorithmic State Machine<br>10. Addressing and Decoding of Memory and I/O Systems |
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| <b>Course Name</b>                      | <b>Logic Circuits and Design Laboratory</b>                                                                                                                                                                                                                                                                |
| <b>Course Description</b>               | <i>This course focuses on providing hands-on experience in designing digital circuits.</i>                                                                                                                                                                                                                 |
| <b>Number of Units for Lecture</b>      | 1 unit                                                                                                                                                                                                                                                                                                     |
| <b>Number of Contact Hours per Week</b> | 3 hours per week                                                                                                                                                                                                                                                                                           |
| <b>Prerequisites</b>                    | Fundamentals of Electronic Circuits                                                                                                                                                                                                                                                                        |
| <b>Co-requisites</b>                    | Logic Circuits and Design                                                                                                                                                                                                                                                                                  |
| <b>Program Outcomes</b>                 | To be identified by the program.                                                                                                                                                                                                                                                                           |
| <b>Course Outcomes</b>                  | To be identified by the program.                                                                                                                                                                                                                                                                           |
| <b>Laboratory Experiments</b>           | <i>Laboratory exercises to be identified by the program.<br/> Each major topic should have a corresponding laboratory exercise.<br/> For semestral program, 15 exercises per semester.<br/> For trimestral program, 12 exercises per trimester.<br/> For quarterterm program, 9 exercises per quarter.</i> |
| <b>Laboratory Equipment</b>             | <i>Program shall provide complete tools and equipment necessary to perform the identified laboratory exercise.<br/> 1 set of tools and equipment per maximum of 5 students per group.</i>                                                                                                                  |

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| <b>Course Name</b>                 | <b>Operating Systems</b>                                                                                                                                                                                                 |
| <b>Course Description</b>          | <i>This course includes different policies and strategies used by an operating system. Topics include operating systems structures, process management, storage management, file management and distributed systems.</i> |
| <b>Number of Units for Lecture</b> | 3 units                                                                                                                                                                                                                  |

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| <b>Number of Contact Hours per Week</b> | 3 hours per week                                                                                                                                                                                                                                                                                                                        |
| <b>Prerequisites</b>                    | Data Structures and Algorithms                                                                                                                                                                                                                                                                                                          |
| <b>Program Outcomes</b>                 | To be identified by the program.                                                                                                                                                                                                                                                                                                        |
| <b>Course Outcomes</b>                  | To be identified by the program.                                                                                                                                                                                                                                                                                                        |
| <b>Course Outline</b>                   | <ol style="list-style-type: none"> <li>1. Overview of the Operating System</li> <li>2. Process Management</li> <li>3. Process Coordination</li> <li>4. Memory Management</li> <li>5. Storage Management</li> <li>6. Protection and Security</li> <li>7. Interfacing to Operating Systems</li> <li>8. Special-Purpose Systems</li> </ol> |

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| <b>Course Name</b>                      | <b>Data and Digital Communications</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Course Description</b>               | This course focuses on the fundamental concepts of digital and data communications. It also includes topics on data security and integrity.                                                                                                                                                                                                                                                                                                                                                               |
| <b>Number of Units for Lecture</b>      | 3 units                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Number of Contact Hours per Week</b> | 3 hours per week                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Prerequisites</b>                    | Fundamentals of Electronic Circuits                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Program Outcomes</b>                 | To be identified by the program.                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Course Outcomes</b>                  | To be identified by the program.                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Course Outline</b>                   | <ol style="list-style-type: none"> <li>1. Elements of Digital Communication</li> <li>2. Pulse Code Modulation</li> <li>3. Digital Modulation Techniques</li> <li>4. Information Theory</li> <li>5. History of Data Communication</li> <li>6. Transmission Media and Transmission Technologies</li> <li>7. Data Transmission Modes and Standards</li> <li>8. Protocols</li> <li>9. Error Detection and Correction</li> <li>10. Encryption and Decryption</li> <li>11. Virus, Worms, And Hacking</li> </ol> |

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| <b>Course Name</b>                      | <b>Introduction to HDL</b>                                                                                                                                                                                                                                                                                                                                         |
| <b>Course Description</b>               | <i>A laboratory course that introduces hardware description language as a tool for designing and testing combinational and sequential circuits. It covers fundamental of concepts of HDL and the basic building blocks of HDL programming.</i>                                                                                                                     |
| <b>Number of Units for Laboratory</b>   | <i>1 unit</i>                                                                                                                                                                                                                                                                                                                                                      |
| <b>Number of Contact Hours per Week</b> | <i>3 hours per week</i>                                                                                                                                                                                                                                                                                                                                            |
| <b>Prerequisites</b>                    | <i>Programming Logic and Design<br/>Fundamentals of Electronic Circuits</i>                                                                                                                                                                                                                                                                                        |
| <b>Program Outcomes</b>                 | <i>To be identified by the program.</i>                                                                                                                                                                                                                                                                                                                            |
| <b>Course Outcomes</b>                  | <i>To be identified by the program.</i>                                                                                                                                                                                                                                                                                                                            |
| <b>Course Outline</b>                   | <ol style="list-style-type: none"> <li>1. Introduction to Hardware Description Language Programming</li> <li>2. Gate Level Modeling</li> <li>3. Dataflow Modeling</li> <li>4. Behavioral Modeling</li> <li>5. Combinational Circuit</li> <li>6. Sequential Circuit</li> <li>7. Counters</li> <li>8. State Machine Design</li> <li>9. Task and Functions</li> </ol> |
| <b>Laboratory Experiments</b>           | <i>Laboratory exercises to be identified by the program.<br/>Each major topic should have a corresponding laboratory exercise.<br/>For semestral program, 15 exercises per semester.<br/>For trimestral program, 12 exercises per trimester.<br/>For quarterterm program, 9 exercises per quarter.</i>                                                             |
| <b>Laboratory Equipment</b>             | <i>Computer and any HDL software tool<br/>Depending on the class size<br/>1 computer per student</i>                                                                                                                                                                                                                                                               |

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| <b>Course Name</b>                      | <b>Feedback and Control Systems</b>                                                                  |
| <b>Course Description</b>               | <i>The course includes the control devices, equations of a systems and block diagram of systems.</i> |
| <b>Number of Units for Lecture</b>      | <i>3 units</i>                                                                                       |
| <b>Number of Contact Hours per Week</b> | <i>3 hours per week</i>                                                                              |

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| <b>Prerequisites</b>    | Numerical Methods<br>Fundamentals of Electrical Circuits                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Program Outcomes</b> | To be identified by the program.                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Course Outcomes</b>  | To be identified by the program.                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Course Outline</b>   | <ol style="list-style-type: none"> <li>1. Introduction to Control System</li> <li>2. Block Diagram Algebra and Transfer Function</li> <li>3. Review of Frequency Response Transfer Function</li> <li>4. Block Diagram of Control Systems</li> <li>5. Types of Feedback</li> <li>6. Frequency Response of Feedback Systems</li> <li>7. Root Locus and Nyquist Criteria</li> <li>8. Stability and Compensation</li> <li>9. Step Response</li> </ol> |

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| <b>Course Name</b>                      | <b>Computer Engineering Drafting and Design</b>                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Course Description</b>               | This course focuses on the principles of layout of electrical, electronics, and logic drawings; stressing modern representation used for block diagrams, wiring/assembly, drawings, printed circuit board layouts, and etching.                                                                                                                                                                                                       |
| <b>Number of Units for Laboratory</b>   | 1 unit                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Number of Contact Hours per Week</b> | 3 hours per week                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Prerequisites</b>                    | Fundamentals of Electronic Circuits                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Program Outcomes</b>                 | To be identified by the program.                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Course Outcomes</b>                  | To be identified by the program.                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Course Outline</b>                   | <ol style="list-style-type: none"> <li>1. Block Diagrams and Flowcharts</li> <li>2. Electrical, Electronic and Logic Components</li> <li>3. Designation, Standards and Abbreviations</li> <li>4. Hand-sketched Schematic Diagrams</li> <li>5. Circuit Layout Simulation Tool</li> <li>6. Wiring and Cabling Diagrams Electronic Packaging</li> <li>7. PCB Design Process</li> <li>8. PCB Design Issues</li> <li>9. Etching</li> </ol> |
| <b>Laboratory Experiments</b>           | Laboratory exercises to be identified by the program.<br>Each major topic should have a corresponding laboratory exercise.<br>For semestral program, 15 exercises per semester.<br>For trimestral program, 12 exercises per trimester.                                                                                                                                                                                                |

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|                             | <i>For quarterterm program, 9 exercises per quarter.</i>                                                                                                                                 |
| <b>Laboratory Equipment</b> | <i>Program shall provide complete tools and equipment necessary to perform the identified laboratory exercise.<br/>1 set of tools and equipment per maximum of 5 students per group.</i> |

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| <b>Course Name</b>                      | <b>Basic Occupational Health and Safety</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Course Description</b>               | <i>This course tackles key Occupational Health and Safety (OSH) concepts, principles and practices that are foundational knowledge requirements applicable in almost all industries. Specifically, it assists learners in identifying the key elements in the OSH situation both here and abroad; determine existing and potential safety and health hazards; identify the range of control measures; discuss pertinent provisions of Philippine laws that refer to occupational safety and health; explain key principles in effectively communicating OSH; identify components of effective OSH programs and demonstrate some skills in identifying hazards and corresponding control measures at the workplace.</i> |
| <b>Number of Units for Lecture</b>      | <i>3 units</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Number of Contact Hours per Week</b> | <i>3 hours per week</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Prerequisites</b>                    | <i>3<sup>rd</sup> Year Standing</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Program Outcomes</b>                 | <i>To be identified by the program.</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Course Outcomes</b>                  | <i>To be identified by the program.</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Course Outline</b>                   | <ol style="list-style-type: none"> <li><i>1. Introductory Concepts</i></li> <li><i>2. Occupational Safety</i></li> <li><i>3. Industrial Hygiene</i></li> <li><i>4. Control Measures for OSH Hazards</i></li> <li><i>5. Occupational Health</i></li> <li><i>6. Personal Protective Equipment</i></li> <li><i>7. OSH Programming</i></li> <li><i>8. Training of Personnel on OSH</i></li> <li><i>9. OSH Legislation</i></li> <li><i>10. Plant Visit Simulation</i></li> </ol>                                                                                                                                                                                                                                            |

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| <b>Course Name</b>                      | <b>Computer Networks and Security</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Course Description</b>               | <i>The course includes the basic principles of network architecture, computer network design, services, technologies and network security.</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Number of Units for Lecture</b>      | 3 units                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Number of Contact Hours per Week</b> | 3 hours per week                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Prerequisites</b>                    | Data and Digital Communications                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Co-requisites</b>                    | Computer Networks and Security Laboratory                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Program Outcomes</b>                 | To be identified by the program.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Course Outcomes</b>                  | To be identified by the program.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Course Outline</b>                   | <ol style="list-style-type: none"> <li>1. Evolution of Computer Networks and Services</li> <li>2. Applications and Layered Architectures</li> <li>3. Local Area Networks (LAN)/Wide Area Networks (WAN) <ol style="list-style-type: none"> <li>3.1 Devices and Protocols</li> <li>3.2 Standards</li> </ol> </li> <li>4. Internetworks <ol style="list-style-type: none"> <li>4.1 Principles of Internetworking</li> <li>4.2 Architectures</li> <li>4.3 IP Addressing and Architecture</li> </ol> </li> <li>5. Network Security <ol style="list-style-type: none"> <li>5.1 Internet Protocol and Standards</li> <li>5.2 Internet Authentication and Applications</li> <li>5.3 Wireless Network Security</li> <li>5.4 Web Security</li> </ol> </li> <li>6. Introduction to Cybersecurity</li> </ol> |

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| <b>Course Name</b>                      | <b>Computer Networks and Security Laboratory</b>                                                                                                                                            |
| <b>Course Description</b>               | <i>This course provides hands-on laboratory activities on computer networking. It focuses on the configuration of TCP/IP, routers and switches, network security and wireless fidelity.</i> |
| <b>Number of Units for Laboratory</b>   | 1 unit                                                                                                                                                                                      |
| <b>Number of Contact Hours per Week</b> | 3 hours per week                                                                                                                                                                            |
| <b>Prerequisites</b>                    | Data and Digital Communications                                                                                                                                                             |
| <b>Co-requisites</b>                    | Computer Networks and Security                                                                                                                                                              |

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| <b>Program Outcomes</b>       | <i>To be identified by the program.</i>                                                                                                                                                                                                                                                                |
| <b>Course Outcomes</b>        | <i>To be identified by the program.</i>                                                                                                                                                                                                                                                                |
| <b>Laboratory Experiments</b> | <i>Laboratory exercises to be identified by the program.<br/>Each major topic should have a corresponding laboratory exercise.<br/>For semestral program, 15 exercises per semester.<br/>For trimestral program, 12 exercises per trimester.<br/>For quarterterm program, 9 exercises per quarter.</i> |
| <b>Laboratory Equipment</b>   | <i>Program shall provide complete tools and equipment necessary to perform the identified laboratory exercise.<br/>1 set of tools and equipment per maximum of 5 students per group.</i>                                                                                                               |

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| <b>Course Name</b>                      | <b>Microprocessors</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Course Description</b>               | <i>This course provides understanding of architecture of microprocessor-based systems; registers, study of microprocessor operation, assembly language, arithmetic operations, and interfacing.</i>                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Number of Units for Lecture</b>      | 3 units                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Number of Contact Hours per Week</b> | 3 hours per week                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Prerequisites</b>                    | Logic Circuits and Design                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Co-requisites</b>                    | Microprocessors Laboratory                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Program Outcomes</b>                 | <i>To be identified by the program.</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Course Outcomes</b>                  | <i>To be identified by the program.</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Course Outline</b>                   | <ol style="list-style-type: none"> <li>1. Structural Components of Microprocessor/Microcontroller <ol style="list-style-type: none"> <li>1.1 Internal CPU Interconnection</li> <li>1.2 ALU</li> <li>1.3 CU</li> <li>1.4 Registers</li> <li>1.5 Other Peripherals</li> </ol> </li> <li>2. Fetch-Decode-Execute Cycle</li> <li>3. Functional Operations of Microprocessor/Microcontroller <ol style="list-style-type: none"> <li>3.1 Data Movement</li> <li>3.2 Data Processing</li> <li>3.3 Control</li> <li>3.4 Data Storage</li> </ol> </li> <li>4. Instruction Set</li> <li>5. I/O Interfacing</li> </ol> |

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|  | 5.1 Interfacing of Input/Output Devices<br>5.2 Interface Devices<br>5.3 Time-Based I/O<br>5.4 Handshaking |
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| <b>Course Name</b>                      | <b>Microprocessors Laboratory</b>                                                                                                                                                                                                                                                                          |
| <b>Course Description</b>               | <i>This course provides understanding of architecture of microprocessor-based systems; study of microprocessor operation, assembly language, arithmetic operations, and interfacing</i>                                                                                                                    |
| <b>Number of Units for Laboratory</b>   | 1 unit                                                                                                                                                                                                                                                                                                     |
| <b>Number of Contact Hours per Week</b> | 3 hours per week                                                                                                                                                                                                                                                                                           |
| <b>Prerequisites</b>                    | Logic Circuits and Design                                                                                                                                                                                                                                                                                  |
| <b>Co-requisites</b>                    | Microprocessors                                                                                                                                                                                                                                                                                            |
| <b>Program Outcomes</b>                 | To be identified by the program.                                                                                                                                                                                                                                                                           |
| <b>Course Outcomes</b>                  | To be identified by the program.                                                                                                                                                                                                                                                                           |
| <b>Laboratory Experiments</b>           | <i>Laboratory exercises to be identified by the program.<br/> Each major topic should have a corresponding laboratory exercise.<br/> For semestral program, 15 exercises per semester.<br/> For trimestral program, 12 exercises per trimester.<br/> For quarterterm program, 9 exercises per quarter.</i> |
| <b>Laboratory Equipment</b>             | Computer and assembly language programming software tool<br>Depending on the class size<br>1 computer per student                                                                                                                                                                                          |

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| <b>Course Name</b>                 | <b>Methods of Research</b>                                                                                                                                                                                                                                                                                 |
| <b>Course Description</b>          | <i>This course will provide in-depth understanding of research through exploration of different research methodologies and ethics. It includes qualitative and quantitative research, descriptive and other applicable research methodologies, inferential statistics and introduction to data mining.</i> |
| <b>Number of Units for Lecture</b> | 2 units                                                                                                                                                                                                                                                                                                    |



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| <b>Number of Contact Hours per Week</b> | 2 hours per week                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Prerequisites</b>                    | Engineering Data Analysis<br>Purposive Communication<br>Logic Circuits and Design                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Program Outcomes</b>                 | To be identified by the program.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Course Outcomes</b>                  | To be identified by the program.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Course Outline</b>                   | <ol style="list-style-type: none"> <li>1. Academic Honesty/Plagiarism</li> <li>2. Types of Research</li> <li>3. Problem Identification</li> <li>4. Literature Search and Review</li> <li>5. Quantitative and Qualitative Methods</li> <li>6. Data Sampling, Collection, and Testing</li> <li>7. Data Analysis and Interpretation</li> <li>8. Validity, Reliability, and Sources of Error</li> <li>9. Citation and Style Mechanics (E.G., APA)</li> <li>10. Article Writing (E.G., IEEE, ACM)</li> <li>11. Presentation and Publication</li> </ol> |

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| <b>Course Name</b>                      | <b>CpE Laws and Professional Practice</b>                                                                                                                                                                                                                                                                                               |
| <b>Course Description</b>               | <i>This course provides the importance of the professional and ethical responsibilities of practicing computer engineers and the effects of their work on society; the importance of understanding contemporary issues, lifelong learning strategies; and applicable IT laws in the field of computer engineering.</i>                  |
| <b>Number of Units for Lecture</b>      | 2 units                                                                                                                                                                                                                                                                                                                                 |
| <b>Number of Contact Hours per Week</b> | 2 hours per week                                                                                                                                                                                                                                                                                                                        |
| <b>Prerequisites</b>                    | 3 <sup>rd</sup> Year Standing                                                                                                                                                                                                                                                                                                           |
| <b>Program Outcomes</b>                 | To be identified by the program.                                                                                                                                                                                                                                                                                                        |
| <b>Course Outcomes</b>                  | To be identified by the program.                                                                                                                                                                                                                                                                                                        |
| <b>Course Outline</b>                   | <ol style="list-style-type: none"> <li>1. Philippine IT Laws and Policies <ol style="list-style-type: none"> <li>a. E-Commerce Law (RA 8792)</li> <li>b. Intellectual Property Code of the Philippines (RA 8293)</li> <li>c. Optical Media Act Of 2003 (RA 9239)</li> <li>d. Data Privacy Act Of 2012 (RA 10173)</li> </ol> </li> </ol> |

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|  | <ul style="list-style-type: none"> <li>e. Department of Information and Communications Technology Act of 2015 (RA 10844)</li> <li>f. Cybercrime Prevention Act of 2012 (RA 10175)</li> <li>2. Philosophical Frameworks and Cultural Issues</li> <li>3. Engineering Solutions and Societal Effects</li> <li>4. Professional and Ethical Responsibilities</li> <li>5. Contemporary Issues</li> <li>6. Lifelong Learning Strategies</li> <li>7. Business and Management Issues</li> <li>8. Tradeoffs in Professional Practice</li> </ul> |
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| <b>Course Name</b>                      | <b>Embedded Systems</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Course Description</b>               | <i>This course provides advanced topics in embedded systems design using contemporary practice; interrupt-driven, reactive, real-time, object-oriented, and distributed client/server embedded systems.</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Number of Units for Lecture</b>      | 3 units                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Number of Contact Hours per Week</b> | 3 hours per week                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Prerequisites</b>                    | Microprocessors                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Co-requisites</b>                    | Embedded Systems Laboratory                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Program Outcomes</b>                 | To be identified by the program.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Course Outcomes</b>                  | To be identified by the program.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Course Outline</b>                   | <ul style="list-style-type: none"> <li>1. History and Overview</li> <li>1. Relevant Tools, Standards, and/or Engineering Constraints</li> <li>2. Characteristics of Embedded Systems</li> <li>3. Basic Software Techniques for Embedded Applications</li> <li>4. Parallel Input and Output</li> <li>5. Asynchronous and Synchronous Serial Communication</li> <li>6. Periodic Interrupts, Waveform Generation, Time Measurement</li> <li>7. Data Acquisition, Control, Sensors, and Actuators</li> <li>8. Implementation Strategies for Complex Embedded Systems</li> <li>9. Techniques for Low-Power Operation</li> <li>10. Mobile and Networked Embedded Systems</li> <li>11. Advanced Topics on Input/Output</li> <li>12. Computing Platforms for Embedded Systems</li> </ul> |

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| <b>Course Name</b> | <b>Embedded Systems Laboratory</b> |
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| <b>Course Description</b>               | <i>This course will provide hands-on activities designed to advanced topics in embedded systems design using contemporary practice; interrupt-driven, reactive, real-time, object- oriented, and distributed client/server embedded systems.</i>                                            |
| <b>Number of Units for Laboratory</b>   | 1 unit                                                                                                                                                                                                                                                                                      |
| <b>Number of Contact Hours per Week</b> | 3 hours per week                                                                                                                                                                                                                                                                            |
| <b>Prerequisites</b>                    | Microprocessors                                                                                                                                                                                                                                                                             |
| <b>Co-requisites</b>                    | Embedded Systems                                                                                                                                                                                                                                                                            |
| <b>Program Outcomes</b>                 | To be identified by the program.                                                                                                                                                                                                                                                            |
| <b>Course Outcomes</b>                  | To be identified by the program.                                                                                                                                                                                                                                                            |
| <b>Laboratory Experiments</b>           | Laboratory exercises to be identified by the program.<br>Each major topic should have a corresponding laboratory exercise.<br>For semestral program, 15 exercises per semester.<br>For trimestral program, 12 exercises per trimester.<br>For quarterterm program, 9 exercises per quarter. |
| <b>Laboratory Equipment</b>             | Program shall provide complete tools and equipment necessary to perform the identified laboratory exercise.<br>1 set of tools and equipment per maximum of 5 students per group.                                                                                                            |

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| <b>Course Name</b>                      | <b>Computer Architecture and Organization</b>                                                                                                                                                                                                                                                                 |
| <b>Course Description</b>               | <i>This course includes the study of the evolution of computer architecture and the factors influencing the design of hardware and software elements of computer systems. The focus is on the understanding of the design issues specifically the instruction set architecture and hardware architecture.</i> |
| <b>Number of Units for Lecture</b>      | 3 units lecture                                                                                                                                                                                                                                                                                               |
| <b>Number of Contact Hours per Week</b> | 3 hours per week                                                                                                                                                                                                                                                                                              |
| <b>Prerequisites</b>                    | Microprocessors                                                                                                                                                                                                                                                                                               |
| <b>Co-requisites</b>                    | Computer Architecture and Organization Laboratory                                                                                                                                                                                                                                                             |
| <b>Program Outcomes</b>                 | To be identified by the program.                                                                                                                                                                                                                                                                              |
| <b>Course Outcomes</b>                  | To be identified by the program.                                                                                                                                                                                                                                                                              |

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| <b>Course Outline</b> | <ol style="list-style-type: none"> <li>1. History and Overview of Computer Architecture</li> <li>2. Relevant Tools, Standards and/or Engineering Constraints</li> <li>3. Instruction Set Architecture</li> <li>4. Measuring Performance</li> <li>5. Computer Arithmetic</li> <li>6. Processor Organization</li> <li>7. Memory System Organization and Architectures</li> <li>8. Input/Output Interfacing and Communication</li> <li>9. Peripheral Subsystems</li> <li>10. Multi/Many-Core Architectures</li> <li>11. Distributed System Architectures</li> </ol> |
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| <b>Course Name</b>                      | <b>Computer Architecture and Organization Laboratory</b>                                                                                                                                                                                                                                                                       |
| <b>Course Description</b>               | <i>This course will provide hands-on activities designed to focus on the computer hardware issues specifically the instruction set architecture and hardware architecture.</i>                                                                                                                                                 |
| <b>Number of Units for Laboratory</b>   | <i>1 unit laboratory</i>                                                                                                                                                                                                                                                                                                       |
| <b>Number of Contact Hours per Week</b> | <i>3 hours per week</i>                                                                                                                                                                                                                                                                                                        |
| <b>Prerequisites</b>                    | <i>Microprocessors</i>                                                                                                                                                                                                                                                                                                         |
| <b>Co-requisites</b>                    | <i>Computer Architecture and Organization</i>                                                                                                                                                                                                                                                                                  |
| <b>Program Outcomes</b>                 | <i>To be identified by the program.</i>                                                                                                                                                                                                                                                                                        |
| <b>Course Outcomes</b>                  | <i>To be identified by the program.</i>                                                                                                                                                                                                                                                                                        |
| <b>Laboratory Experiments</b>           | <i>Laboratory exercises to be identified by the program.</i><br><i>Each major topic should have a corresponding laboratory exercise.</i><br><i>For semestral program, 15 exercises per semester.</i><br><i>For trimestral program, 12 exercises per trimester.</i><br><i>For quarterterm program, 9 exercises per quarter.</i> |
| <b>Laboratory Equipment</b>             | <i>Program shall provide complete tools and equipment necessary to perform the identified laboratory exercise.</i><br><i>1 set of tools and equipment per maximum of 5 students per group.</i>                                                                                                                                 |

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| <b>Course Name</b>        | <b>Emerging Technologies in CpE</b>                                                                                                                 |
| <b>Course Description</b> | <i>This course is designed to provide flexibility in the curriculum by discussing any emerging technologies applicable to computer engineering.</i> |

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| <b>Number of Units for Lecture</b>      | 3 units                                           |
| <b>Number of Contact Hours per Week</b> | 3 hours per week                                  |
| <b>Prerequisites</b>                    | 4 <sup>th</sup> Year Standing                     |
| <b>Program Outcomes</b>                 | To be identified by the program.                  |
| <b>Course Outcomes</b>                  | To be identified by the program.                  |
| <b>Course Outline</b>                   | Depending on the topic chosen by the institution. |

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| <b>Course Name</b>                      | <b>Seminars and Fieldtrips</b>                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Course Description</b>               | The course includes seminars and lectures on current trends and issues on Computer Engineering developments. Include field trips to different companies and plants dealing with computer system facilities.                                                                                                                                                                                            |
| <b>Number of Units for Laboratory</b>   | 1 unit                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Number of Contact Hours per Week</b> | 3 hours per week                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Prerequisites</b>                    | 4 <sup>th</sup> Year Standing                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Program Outcomes</b>                 | To be identified by the program.                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Course Outcomes</b>                  | To be identified by the program.                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Course Outline</b>                   | <ol style="list-style-type: none"> <li>1. Seminars <ol style="list-style-type: none"> <li>1.1 Technical Seminars (Minimum of 3)</li> <li>1.2 Non-Technical Seminars <ol style="list-style-type: none"> <li>1.2.1 Career Development</li> <li>1.2.2 Labor Education</li> </ol> </li> </ol> </li> <li>2. Fieldtrips (Minimum of 2 Company Visits)</li> <li>3. Submission of Student Portfolio</li> </ol> |

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| <b>Course Name</b>        | <b>Digital Signal Processing</b>                                                                                                                                                                               |
| <b>Course Description</b> | The course includes the need for and tradeoffs made when sampling and quantizing a signal; linear, time-invariant system properties; frequency as an analysis domain complementary to time; and filter design. |

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| <b>Number of Units for Lecture</b>      | 3 units                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Number of Contact Hours per Week</b> | 3 hours per week                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Prerequisites</b>                    | Feedback and Control Systems                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Co-requisites</b>                    | Digital Signal Processing Laboratory                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Program Outcomes</b>                 | To be identified by the program.                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Course Outcomes</b>                  | To be identified by the program.                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Course Outline</b>                   | <ol style="list-style-type: none"> <li>1. History and Overview</li> <li>2. Relevant Tools, Standards, and/or Engineering Constraints</li> <li>3. Convolution</li> <li>4. Transform Analysis</li> <li>5. Frequency Response</li> <li>6. Sampling and Aliasing</li> <li>7. Digital Spectra and Discrete Transforms</li> <li>8. Finite and Infinite Impulse Response Filter Design</li> <li>9. Window Functions</li> <li>10. Multimedia Processing</li> </ol> |

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| <b>Course Name</b>                      | <b>Digital Signal Processing Laboratory</b>                                                                                                                                                                                                                                                                |
| <b>Course Description</b>               | <i>This course is designed to provide hands-on activities on different applications of digital signals processing.</i>                                                                                                                                                                                     |
| <b>Number of Units for Laboratory</b>   | 1 unit                                                                                                                                                                                                                                                                                                     |
| <b>Number of Contact Hours per Week</b> | 3 hours                                                                                                                                                                                                                                                                                                    |
| <b>Prerequisites</b>                    | Feedback and Control Systems                                                                                                                                                                                                                                                                               |
| <b>Co-requisites</b>                    | Digital Signal Processing                                                                                                                                                                                                                                                                                  |
| <b>Program Outcomes</b>                 | To be identified by the program.                                                                                                                                                                                                                                                                           |
| <b>Course Outcomes</b>                  | To be identified by the program.                                                                                                                                                                                                                                                                           |
| <b>Laboratory Experiments</b>           | <i>Laboratory exercises to be identified by the program.<br/> Each major topic should have a corresponding laboratory exercise.<br/> For semestral program, 15 exercises per semester.<br/> For trimestral program, 12 exercises per trimester.<br/> For quarterterm program, 9 exercises per quarter.</i> |

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| <b>Laboratory Equipment</b> | <i>Computer and DSP software tool<br/>Depending on the class size<br/>1 computer per student</i> |
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| <b>Course Name</b>                      | <b>CpE Practice and Design 1</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Course Description</b>               | <i>This course is the first course in a two-semester sequence that constitutes the design experience for undergraduate computer engineers. It provides essential ideas, concepts and principles in engineering design process and emphasizes other design issues including engineering standards and multiple constraints as well as effective communication strategies. Students work in teams to develop project proposals for assigned open-ended problems. Students are required to make oral presentations and submit written proposal for their projects.</i> |
| <b>Number of Units for Laboratory</b>   | <i>1 unit</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Number of Contact Hours per Week</b> | <i>3 hours per week</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Prerequisites</b>                    | <i>Microprocessors<br/>Methods of Research</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Program Outcomes</b>                 | <i>To be identified by the program.</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Course Outcomes</b>                  | <i>To be identified by the program.</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Course Outline</b>                   | <ol style="list-style-type: none"> <li><i>1. Orientation</i></li> <li><i>2. Relevant Tools, Standards, and/or Engineering Constraints</i></li> <li><i>3. Effective Communication Strategies</i></li> <li><i>4. Intellectual Property and Legal Issues</i></li> <li><i>5. Submission of Design Proposal</i></li> <li><i>6. Presentation of Design Proposal</i></li> <li><i>7. Submission of Approved Proposal</i></li> </ol>                                                                                                                                         |
| <b>Laboratory Equipment</b>             | <i>Computer and any programming language and/or simulation software tool; materials, components and tools needed for prototype development and testing.</i>                                                                                                                                                                                                                                                                                                                                                                                                         |



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| <b>Course Name</b>                      | <b>CpE Practice and Design 2</b>                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Course Description</b>               | <i>This course is the second of the design experience for undergraduate computer engineering students. In this course, students will be expected to build/fabricate their design, test and evaluate the design against their design specifications, and demonstrate a fully functional project to their design review committee. Students make oral presentations and submit final reports documenting their projects.</i> |
| <b>Number of Units for Laboratory</b>   | 2 units                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Number of Contact Hours per Week</b> | 6 hours per week                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Prerequisites</b>                    | CpE Practice and Design 1                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Program Outcomes</b>                 | To be identified by the program.                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Course Outcomes</b>                  | To be identified by the program.                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Course Outline</b>                   | <ol style="list-style-type: none"> <li>1. Orientation</li> <li>2. Final oral presentation</li> <li>3. Submission of final document</li> </ol>                                                                                                                                                                                                                                                                              |
| <b>Laboratory Equipment</b>             | Computer and any programming language and/or simulation software tool; materials, components and tools needed for prototype development and testing.                                                                                                                                                                                                                                                                       |

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| <b>Course Name</b>                      | <b>On the Job Training</b>                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Course Description</b>               | <i>This course enables students to relate their acquired competencies to the realities and problems of industries in a multidisciplinary environment. This may include involvement in the industry's manpower requirements, development and research concerns, trainings, applications of principles, environmental concerns, ethical and behavioral concerns, decision making, and equipment and materials concerns.</i> |
| <b>Number of Units for Lecture</b>      | 3 units                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Number of Contact Hours per Week</b> | 3 hours per week                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Minimum Number of Hours</b>          | 240 hours of field work                                                                                                                                                                                                                                                                                                                                                                                                   |

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| <b>Required for Field Work</b> |                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Prerequisites</b>           | <i>4th Year Standing</i>                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Program Outcomes</b>        | <i>To be identified by the program.</i>                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Course Outcomes</b>         | <i>To be identified by the program.</i>                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Course Outline</b>          | <ol style="list-style-type: none"> <li>1. <i>Orientation and Presentation of Policies and Guidelines</i></li> <li>2. <i>Multidisciplinary Team Approaches</i></li> <li>3. <i>Assertion of Student's OJT on the Company</i></li> <li>4. <i>Completion of 240 Hours</i></li> <li>5. <i>Submission of Progress Reports</i></li> <li>6. <i>Final Oral Presentation</i></li> <li>7. <i>Submission of Final Report</i></li> </ol> |